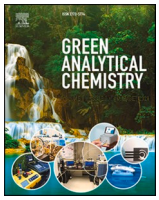
Green Analytical Chemistry 12 (2025) 100214



Contents lists available at ScienceDirect

Green Analytical Chemistry

journal homepage: www.elsevier.com/locate/greeac





Liquid chromatography-high resolution mass spectrometry based analysis of persistent mobile organic compounds in aqueous samples: Method development and optimization

Kyriaki Anagnostopoulou [a,b], Vasileios Alampanos [a,b], Eleni Evgenidou [a,b], Dimitra A. Lambropoulou [a,b,*]

[a] Department of Chemistry, Aristotle University of Thessaloniki, GR 54124, Thessaloniki, Greece
[b] Centre for Interdisciplinary Research and Innovation (CIRI-AUTH), Balkan Center, Thessaloniki, 10th km Thessaloniki-Thermi Rd, GR 57001, Greece

ARTICLE INFO

Keywords:
Potentially persistent
Mobile
Organic compounds
Method development
HILIC
Central composite design (CCD)

ABSTRACT

Persistent Mobile Organic Compounds (PMOCs) represent a critical threat to water resources due to their high solubility, resistance to degradation, and potential toxicity, posing risks to human health and ecosystems. Their detection and analysis are particularly challenging, necessitating advanced analytical techniques to support regulatory measures and ensure safe drinking water. This study addresses these challenges by developing and optimizing a robust method for PMOC detection, integrating solid-phase extraction with liquid chromatography-high resolution mass spectrometry (SPE-LC-HRMS). By bridging analytical gaps in PMOC identification, the research contributes to water quality protection and aligns with global environmental and public health goals. The method is designed to accommodate the diverse physicochemical properties of PMOCs, enabling the detection of a broad spectrum of compounds. Innovations include the creation of a comprehensive PMOC target list, optimization of sample preparation with specialized SPE cartridges, and the fine-tuning of chromatographic and HRMS parameters. A central composite design and desirability function (DF) approach were used to optimize the HRMS analysis, while the most suitable SPE cartridge, column, and mobile phase composition were carefully selected through experimental trials. The method achieved recoveries within 70–120 % and excellent linearity ($r^2 > 0.997$) for all analytes. Validated under European directives (2002/657/EC and SANTE 11,813/2017), the study offers a sensitive, reliable, and regulatory-compliant framework for PMOC monitoring in water systems. By addressing both scientific and regulatory demands, this work advances environmental analytical chemistry and strengthens efforts to protect water resources from emerging contaminants.

1. Introduction

Protecting drinking water supplies from hazardous and undesired substances is essential to upholding human rights and preserving public health [1]. This principle is a matter of great interest for many governments and organizations. For example, it is firmly embedded in European policies and directives, such as the EU Water Framework Directive and the Drinking Water Directive, which aim to ensure high standards of water quality across member states. Effective policies, regulations, and community engagement are crucial in maintaining water quality and safeguarding this vital resource for present and future generations. These directives support the Sustainable Development Goals (SDGs), particularly Goal 6, which focuses on clean water and sanitation, aiming to improve water quality through pollution reduction [2].

Physicochemical characterization is vital for understanding chemical behavior in water and supporting environmental directives. As of January 2023, CAS registered 219 million chemicals, and PubChem listed 116 million, offering extensive resources for chemical assessments [1]. To better understand the behavior and fate of chemicals in water, key traits such as aquatic persistency (P) and mobility (M) are crucial, as they enable chemicals, known as PMOCs, to potentially pollute drinking water by traversing natural and man-made barriers like aquifers and riverbanks [1]. Moreover, a subset of these substances can also be classified as very persistent and very mobile (vPvM) or persistent, mobile, and toxic (PMT/vPvM) compounds. The criteria for PMT/vPvM

* Corresponding author.
E-mail address: dlambro@chem.auth.gr (D.A. Lambropoulou).

https://doi.org/10.1016/j.greeac.2025.100214
Received 30 December 2024; Received in revised form 25 January 2025; Accepted 28 January 2025
Available online 29 January 2025
2772-5774/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

compounds are outlined in the EU Chemicals, Labelling and Packaging (CLP) regulation (EC 1272/2008) [3]. vPvM substances are defined as those that "Can cause very long-lasting and diffuse contamination of water resources," whereas PMT substances are defined as those that "Can cause long-lasting and diffuse contamination of water resources" under the Classification, Labelling, and Packaging regulation substances [4]. The Scientific Committee on Health, Environmental and Emerging Risks (SCHEER) of the European Commission has identified PMT/vPvM compounds as one of the 14 emerging issues that potentially have an effect on human health or the environment [2]. From a legal standpoint, a regulatory framework under REACH was established based on the risk posed by mobile and persistent compounds, which was initially brought up by the German Environment Agency (UBA) in 2009. Finally, in April of 2023, the CLP regulation brought into effect the new hazard classes, PMT and vPvM [5].

Following their recognition, a growing number of publications pertaining to the analysis of the substances mentioned above have been released. The liquid chromatography coupled with high-resolution mass spectrometry aspires to be a promising technique for the quantification of PMOCs, based on their unique physicoochemical characteristics. Based on a thorough literature review many analytic methods for PMOC identification have been published. To begin with, for the quantitative identification of 23 target PMOCs, Montes et al., 2019 suggested a method combining mixed-mode-solid-phase extraction (MM-SPE) as a preconcentration approach and mixed-mode liquid chromatography (MMLC) coupled to tandem mass spectrometry [6]. Furthermore, a method for determining 1310 possible PMOCs by supercritical fluid chromatography (SFC) and hydrophilic interaction liquid chromatography (HILIC) linked to high resolution mass spectrometry (HRMS) was proposed by Neuwald et al. [7]. Moreover, Minkus et al., 2020 aim to make a wide range of PMOCs analytically accessible by serially connecting hydrophilic interaction liquid chromatography (HILIC) and reversed-phase liquid chromatography to high-resolution mass spectrometry (HRMS) [8].

The ultimate goal of this study was to narrow the knowledge gap in the analysis and as a result, the occurrence of (potential) PMOCs compounds in the aquatic environment, by establishing a new carefully designed and optimized SPE-LC-HRMS method. The first objective was to develop tools for effective target analysis of these compounds by generating a comprehensive target list of PMOCs candidates from diverse sources, based on available standards. This target list was then combined with optimized analytical techniques, including sample preparation using various extraction materials—hydrophilic-lipophilic balance polymer (HLB), weak anion exchange (WAX), and mixed-mode polymeric sorbent (MCX)—two mobile phases, and two chromatographic methods (HILIC and RPLC). Finally, a Central Composite Design (CCD) approach was employed to determine the optimal mass spectrometry (MS) parameters for the accurate quantification of potential PMOCs. The integration of these techniques aimed to enhance both the sensitivity and selectivity required for the reliable detection and quantification of PMOCs in water samples.

2. Materials and methods

2.1. Chemicals standards and, reagents

All solvents employed in the current study, specifically methanol (MeOH) and water, were of LC-MS grade. Formic acid (FA) was procured from Fisher Scientific (Waltham, MA, USA), while ammonium formate (NH_4OFA) and ammonium hydroxide were sourced from Sigma-Aldrich (Steinheim, Germany). Pesticide standards were acquired through Restek (Bellefonte, USA) multiresidue kits, while standards for pharmaceuticals and personal care products (PPCPs) were of high analytical grade (>95 %) and supplied by Sigma Aldrich, Fluka, and Dr. Ehrenstorfer from Wellington Laboratories (Ontario, Canada).

A comprehensive inventory detailing the target compounds, inclusive of CAS numbers and principal properties, is presented in Table S1. Fresh working solutions were meticulously prepared on each operational day by dilution of the stock in methanol/water (90/10 v/v). Extraction cartridges, specifically hydrophilic-lipophilic balance polymer (HLB, 200 mg/6 mL), weak anion exchange (WAX, 150 mg/6 mL), and mixed-mode polymeric sorbent (MCX 150 mg/6 mL), branded by Oasis, were procured from Waters Corporation (Milford, MA).

2.2. Sample preparation workflow

Solid Phase Extraction (SPE) offers numerous advantages in sample preparation and analytical processes, significantly enhancing the efficiency, sensitivity, and reliability of analytical results. One of the primary benefits of SPE is its ability to concentrate from large volumes of samples, thus improving the detection limits of analytical methods. This is particularly crucial in trace analysis of contaminants in environmental samples. Additionally, SPE provides excellent selectivity by employing a variety of sorbent materials, which are easily accessible for each laboratory, tailored to target specific compounds, which reduces matrix interferences and enhances the purity of the extracts. An SPE method is highly reproducible and can be automated, leading to increased throughput and consistency in routine analyses. Furthermore, the versatility of SPE allows it to be used for a wide range of applications, including drug monitoring, forensic analysis, and food safety testing. Overall, SPE is a powerful tool that streamlines the sample preparation process, providing cleaner samples and more accurate analytical results [9].

In summary, following the optimized procedure, aqueous samples (500 mL of tap-drinking water) were spiked with the solution with target compounds to achieve a concentration of 100 ng L^{-1}, and adequately stirred. Simultaneously, WAX cartridges underwent conditioning with 4 mL of 0.1 % ammonia in MeOH, followed by 4 mL of MeOH, and then 4 mL of water. Similarly, the HLB cartridges were conditioned with 5 mL of water and 5 mL of methanol, while the MCX cartridges with 6 mL of MeOH, 3 mL of water and 3 mL with water with pH adjusted at 2.5. Subsequently, with the sorbent consistently moist, the samples were loaded onto the cartridges at a flow rate of 1–2 mL min^{-1}, facilitated by polypropylene (PP) tubes (60 mL). Upon complete percolation of the entire sample, WAX and HLB cartridges were rinsed successively with 5 mL of water and MCX cartridges with 5 mL of acidified water, to eliminate any potential interferences within the cartridge, concluding with a vacuum-drying process lasting 30 min. Finally, the analytes were eluted into PP conical tubes using 4 mL of MeOH and 4 mL of 0.1 % ammonia in MeOH for WAX cartridges, 7 mL of MeOH for HLB, and with 8 mL of 2 % ammonia in MeOH for MCX cartridges. The resulting extracts were evaporated close to dryness under a gentle stream of nitrogen at 30 °C using a Techne sample concentrator by Cole-Parmer (Staffordshire, UK), reconstituted with 0.5 mL of MeOH/water (90/10, v/v) (yielding an enrichment factor of 1000), and subsequently filtered through PTFE 0.22 μm filters before being introduced into the LC–Orbitrap-MS system.

To begin with, divinylbenzene, a lipophilic compound, and N-vinylpyrrolidone, a hydrophilic compound, copolymerize to form HLB cartridges, designed to have both hydrophilic and lipophilic properties. This combination allows the cartridges to retain a wide range of chemicals from highly polar to non-polar through hydrophobic, π-π, and hydrogen bonding interactions. This balance makes HLB cartridges versatile and effective for extracting diverse compounds from complex sample matrices. The carbonyl oxygen and the nitrogen in the pyrrolidone ring serve as the sorbent's primary hydrogen bond acceptors. Each N-vinylpyrrolidone unit typically has two hydrogen bond acceptors. The divinylbenzene component provides hydrophobic character through its aromatic rings, facilitating interactions with non-polar analytes via hydrophobic forces and van der Waals interactions. HLB cartridges are especially versatile due to their capacity to handle a wide range of pKa values and their resilience across an extensive pH range (pH 0–14). Moreover, HLB cartridges are versatile, retaining both polar and non-

polar molecules, making them suitable for various pKa values and complex sample matrices [10].

Acidic substances with a range of pKa values can be extracted using Weak Anion Exchange (WAX) cartridges. In general, analytes with pKa values between two and eight perform well with WAX cartridges. The primary interaction in WAX cartridges is ionic. The anion exchange sites on the cartridge are functional groups that attract and retain negatively charged species (anions) from the sample. Acidic compounds with negatively charged functional groups (e.g., carboxylate, phosphate, sulfonate) form ionic bonds with the positively charged sites on the WAX cartridge. Additionally, hydrogen bonds can form between the analytes and the functional groups on the cartridge if there are suitable donor and acceptor groups present. Although secondary to ionic interactions, hydrogen bonding can improve the retention of some molecules. In order to improve sample preparation adaptability, the sorbent in WAX cartridges typically includes both anion exchange capabilities and frequently hydrophobic properties. Many WAX sorbents are silica-based, providing a robust and stable backbone. Silica particles are often modified to introduce the desired functional groups while some sorbents are made from polymeric materials, which can offer higher pH stability, and a broader range of applications compared to silica-based sorbents.

On the other hand, basic compounds are extracted using MCX cartridges. The analytes' pKa plays a critical role in evaluating whether MCX cartridges are appropriate and effective for a given extraction. MCX cartridges are designed to retain basic substances by establishing ionic connections with negatively charged functional groups on the resin. Its copolymer of divinylbenzene and sulfonated styrene, known as a mixed-mode sorbent, combines the features of reversed-phase and cation-exchange. Divinylbenzene provides the hydrophobic backbone for reversed-phase interactions, while sulfonated styrene contains sulfonic acid groups (-SO3H) that provide cation-exchange properties (the hydrogen atom in the sulfonic acid group can act as a hydrogen bond donor and oxygen atoms in the sulfonic acid group can act as hydrogen bond acceptors due to their lone pairs of electrons) [10]. Typically, analytes having pKa values between roughly 7 and 12 can use MCX cartridges for their extraction. This range guarantees that the analytes can be neutralized during the elution phase and fully protonated (positively charged) during the loading phase. However, maintaining a high enough pH to ensure full protonation for analytes with pKa values less than 7 may be difficult without risking solvent and SPE cartridge compatibility issues. Conversely, for analytes with pKa values higher than 12, it may be challenging to completely protonate the basic groups, potentially diminishing the efficiency of the interaction with the MCX cartridge [11,12]. In addition to pKa values, the octanol-water partition coefficient (logKow) should be considered when explaining analyte recovery. LogKow measures a chemical's hydrophobicity by indicating how it partitions between hydrophilic (water) and hydrophobic (octanol) phases, and this characteristic plays a significant role in the SPE process. For both WAX and MCX cartridges, higher logKow values indicate increased hydrophobicity, resulting in stronger hydrophobic interactions with the sorbent material. This enhances the retention of these compounds, but it may also complicate elution, requiring stronger solvents or harsher conditions to break both ionic and hydrophobic interactions. Conversely, low logKow compounds, being more hydrophilic, rely primarily on ionic interactions for retention, and their reduced hydrophobicity may lead to weaker overall retention [13,14].

2.3. LC-MS/MS system operation

A Q Exactive TM Focus Orbitrap LC-MS/MS System by Thermo Fisher Scientific, (Bremen, Germany), was employed for the instrumental analysis. The liquid chromatography (LC) component consisted of a Dionex Ultimate 3000 system, encompassing a binary gradient ultra-high-performance liquid chromatography (UHPLC) system with pumps, a temperature-controlled autosampler, and a column compartment. The separation of analytes was conducted using two different columns, one Thermo Hypersil GOLD aQ column (50 × 2.1 mm, 1.9 μm) equipped with a holder and a cartridge pre-filter and one HILIC (2.1 × 100 mm, 1.7 μm), both maintained at a constant temperature of 40 °C throughout the analysis.

The mobile phases utilized in the chromatographic separation can be categorized in two different mixture: water (A) and methanol (B), both containing 0.1 % (v/v) formic acid (FA); the second is water (A) and acetonitrile (B), both containing 0.1 % (v/v) formic acid (FA); the third is water (A) and acetonitrile (B), both containing 0.1 % (v/v) acetic acid (AA); and finally, the fourth is water (A) and methanol (B), both containing 5 mM ammonium formate. The mobile phase gradient program for both solvent mixtures was initiated at 10 % mobile phase B, sustained for 1.5 mins; subsequently, the proportion of mobile phase B increased to 60 % over the following 2.5 mins and further increased to 70 % over the subsequent 4 mins, ultimately reaching 100 % at the 11-minute mark and remaining constant for an additional 1 min. Subsequently, the mobile phase was reverted to the initial conditions within a minute and maintained for 2 mins to facilitate re-equilibration. The total duration of the analytical procedure was 15 mins. The flow rate of the mobile phase was consistently maintained at 200 μL/min throughout the run. The injection volume was fixed at 5 μL, and the autosampler tray temperature was stabilized at 10 °C.

As far as the mass spectrometer operation is concerned, a heated electrospray ionization (HESI-II) probe, specifically the Ion Max variant, was employed for ionization in both positive and negative modes. In addition to the ion source, critical components of the mass spectrometer included the quadrupole mass filter, the high collision dissociation (HCD) cell, the C-trap, and the Orbitrap mass analyzer, facilitating high-resolution and high-mass-accuracy analysis. The HESI parameters were configured in accordance with system auto-defaults for an LC flow rate of 0.2 mL min^{-1}, delineated as follows: tube lens at 110 V; flow rates for sheath, auxiliary gas, and sweep gas at 45, 10, and 2 arbitrary units (au) respectively, sourced from a high-purity generator (Peak Scientific Genius 1022, Inchinnan, UK); spray voltage at 3.5 kV in positive ESI and 2.5 kV in negative ESI; S-lens RF level at 50. Finally, the auxiliary gas heater temperature was tested at 200, 300 and 400 °C, the capillary temperature was tested at 280, 300, 320 °C and the probe at B, C, and D positions.

For quantitative analysis, full-scan (FS) mode acquisition was implemented within a scan range of 100–1000 *m/z*, attaining a resolution of 70,000 (FWHM at 200 Da). The automatic gain control (AGC) was configured at 10^6, and the maximum inject time (IT) was set to auto. Concurrently, a ddMS2 experiment was conducted to facilitate unambiguous identification and confirmation in a single injection. MS2 spectra were acquired solely for precursor ions specified in the user-created inclusion list, utilizing stepped collision energies of 15, 30, and 45 eV for fragmentation, at a resolution of 17,500. Other ddMS2 parameters included AGC at 2×10^5 and IT set to auto. The apex trigger feature was activated and set at 2–5 s to enhance fragmentation sensitivity, with an isolation window of 1.0 *m/z*. The recorded data type for spectra was centroid in both instances.

At the initiation of each sequence, solvent blanks mirroring the solvent composition of the final extracts were introduced three times. These solvent blanks were administered either after every five samples or interleaved between individual samples of authentic specimens, aiming to preclude potential cross-contamination and evaluate any conceivable carryover effect. Additionally, a blend of representative compounds encompassing the designated mass range (referred to as the system suitability solution) was introduced at concentrations of 50 μg/L at the sequence's outset, midpoint, and conclusion to assess the overall integrity of the analytical system, based on the generated analytical response. Quality control samples (QC) were systematically prepared, extracted, and introduced at the commencement and termination of each analysis, ensuring the comprehensive integrity of the analytical process and serving as a corrective measure for any potential analytical inconsistencies.

2.4. Method validation

The validation of the methodology, along with the implementation of quality assurance and quality control procedures (QA/QC), adhered to the guidelines outlined in DG SANTE 12,682/2019 [15] and ISO/IEC 17,025:2017. The assessment of instrumental performance involved the determination of instrument limits, ascertained through ten consecutive injections, and the evaluation of linearity. Comprehensive validation of the overall methodology encompassed the examination of method detection and quantification limits (MDL, MQL), recoveries (%R), precision (expressed as relative standard deviation, RSD), and linearity. Repeatability (%RSDr) was assessed by analyzing five replicates at a spiking level of 50 ng/L within a single day, while within laboratory reproducibility (%RSDWR) was evaluated over five consecutive days. Recoveries were computed as the mean of five replicates ($n = 5$). To mitigate potential contamination in the spiked blanks, additional blank analyses were routinely conducted to subtract the area of any detectable peak at quantifying levels.

2.5. Identification criteria

The identification of analytes adhered to the most recent iteration of DG SANTE 12,682/2019 [16]. All recommendations provided by the DG were meticulously examined, and the pertinent constraints were integrated into the Trace Finder Quan Method. In succinct terms, positive identification was established based on (a) the discernible presence of the adduct ion peak in the extracted ion chromatogram, exhibiting a permissible retention time (t_R) drift of < 0.1 min in comparison to the (matrix) standard; (b) a mass error of < 5 ppm for the adduct ion in the full-scan spectrum; (c) the isotopes' cluster in the full-scan spectrum with a mass error < 5 ppm and a relative isotope abundance (RIA) surpassing a score threshold of > 70 %; (d) the existence of at least one characteristic fragment in the MS2 spectrum with a mass error below 5 ppm; and (e) an adequate signal-to-noise ratio (S/N) for the quantification peak >10. Considering background noise in HRMS equipment is hard to discern, signals lower than 10^4 were regarded as noise and not further as positive findings

2.6. Design of experiments for HRMS optimization

Optimizing the ion source of an LC-MS system is a crucial first step in method development. This enables the user to alter vital instrument parameters to achieve maximum sensitivity and robustness. A LC-MS system, such as the Orbitrap (Q Exactive Focus), uses tools to automatically set standard conditions in order to accomplish this operation.

The experimental design, response processing, and statistical significance assessment of the equations for optimizing High-Resolution Mass Spectrometry analysis were conducted using the Design-Expert software (Trial version 7.0.0, Stat-Ease, Inc., Minneapolis, USA). Consequently, a Design of Experiments (DoE) methodology was adopted to achieve statistical optimization. A CCD approach consisting of three factors was employed among the available possibilities.

The experimental variables included: A) Auxiliary temperature (200, 300, 400 °C), B) capillary temperature (280, 300, 320 °C), and C) probe position [B, C, D (distance from the source)]. Optimization procedures involved examining the responses derived from a statistically designed combination, determining coefficients through fitting experimental data to response functions, and predicting responses from the fitted model. Various diagnostic tests were applied to ensure the adequacy of the fit, including assessments for the presence of outliers, normal probability plots, Cook's distance, leverage, graphs depicting residuals versus predicted values, and graphical representations of predicted versus experimental values. Analysis of variance (ANOVA) at a 95 % confidence level served as the appropriate statistical tool for assessing the significance of factors. Optimal conditions for response variables were illustrated using response surfaces and contour plots. Numerical optimization within the software was employed to determine the specific point maximizing the desirability function. Each parameter was set to a desired goal (maximize, minimize, target, within range, or none), accompanied by upper and lower limits for each variable, considering equal relative importance. Numerical optimization prioritized the recovery (%) and its value as 'maximize' for the goal of the highest importance.

3. Results and discussion

3.1. Selection of the optimum SPE cartridge

The SPE protocol was optimized along with a previously developed HRMS-based multi-residue method by the authors, which encompassed the majority of target compounds. In general, SPE operates through an exhaustive extraction mode where kinetic factors have little influence on extraction efficiency. In contrast, thermodynamic factors play a critical role. Unequivocally, the most critical factor is the sorbent material, which determines the type of intermolecular interactions and is directly related to the mass of analytes extracted. Three different types of cartridges were tested to achieve the best results regarding the method recoveries. Specifically, HLB, WAX, and MCX cartridges from Waters Corporation (Milford, MA) were evaluated.

In an nutshell, HLB, WAX, and MCX cartridges were selected for their ability to extract a diverse range of analytes with varying properties. HLB cartridges retain both polar and non-polar compounds through hydrophobic, π-π, and hydrogen bonding interactions, making them highly versatile across a wide pH range (0–14). WAX cartridges are ideal for acidic compounds, relying on ionic interactions with negatively charged analytes and supporting hydrophobic retention for compounds with pKa values between 2 and 8. MCX cartridges specialize in extracting basic compounds through a combination of reversed-phase and cation-exchange interactions, effectively retaining protonated analytes with pKa values between 7 and 12. Together, these cartridges ensure efficient extraction of compounds with diverse polarity and pKa, as it was mentioned in previous section.

Comprehensive protocols are delineated in Fig. 1. To assess the efficacy of each protocol, the recovery of target analytes (Fig. 2) was calculated following the introduction of spiked samples at one concentration value. As highlighted in the figure, HLB cartridges present the highest recoveries for the majority of the analyzed compounds. As it was also mentioned in the literature, the HLB phase is a useful SPE cartridge for broad metabolite recovery and sample cleanup [16]. Nevertheless, certain significant chemicals displayed better recovery rates using MCX or WAX cartridges. Specifically, prednisone, fluconazole, venlafaxine, irbersasrtan and losartan exhibits their highest recovery (>70 %) by utilizing the MCX based procedure. This is supported by the fact that all of them, based on their chemical properties (Table S2, pKa), exhibit basic characteristics and as a result MCX are the most effective approach for extracting basic compounds. On the other hand, alprazolam, a well-known acidic compound, requires WAX based pretreatment for better extraction recovery. However, the recoveries using HLB cartridges are between the recommended limits according to ISO/IEC 17, 025:2017, as well.

In summary, the three SPE cartridges shown in Fig. 2 were developed using the mixed mode anion and cation exchange phases as well as hydrophilic-lipophilic interactions to achieve better recovery for the PMOC study. When the HLB cartridges were employed, the majority of the compounds had recovery rates between 70 % and 100 %, as seen in Fig. 3A. Nonetheless, certain chemicals may exhibit superior extraction using the WAX or MCX based process due to their specific properties; this does not imply that the HLB extractions yielded poor recoveries, but rather that some compounds respond better to other sorbent types. Concluding, HLB were chosen as the appropriate cartridge for the extraction of PMOCs in this study.

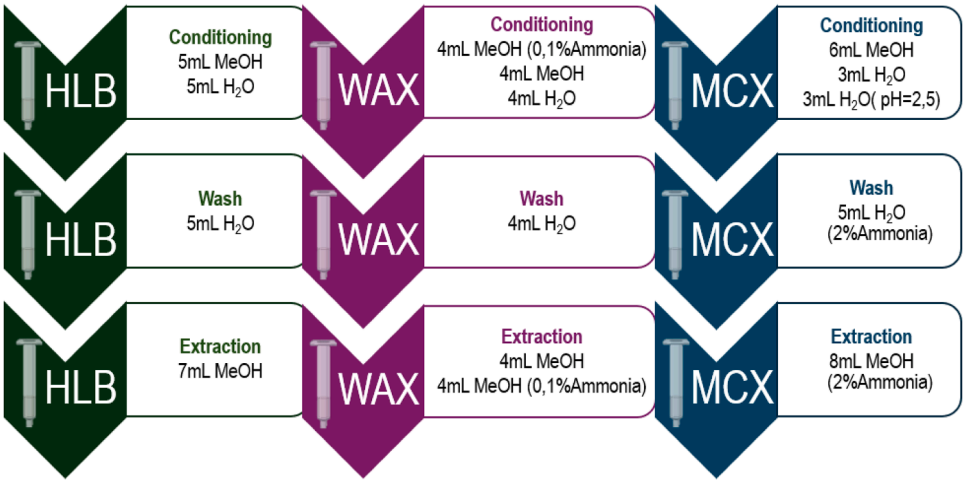
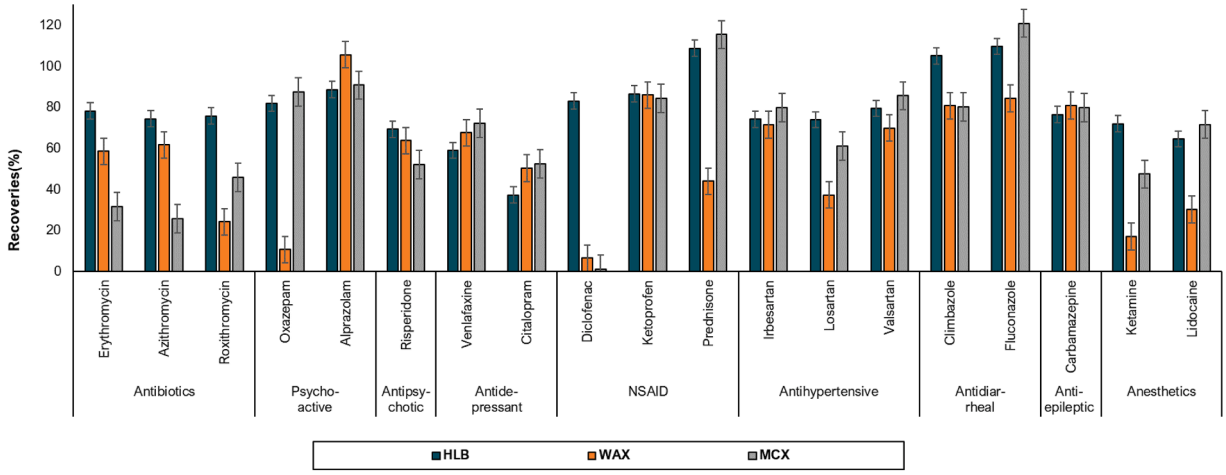
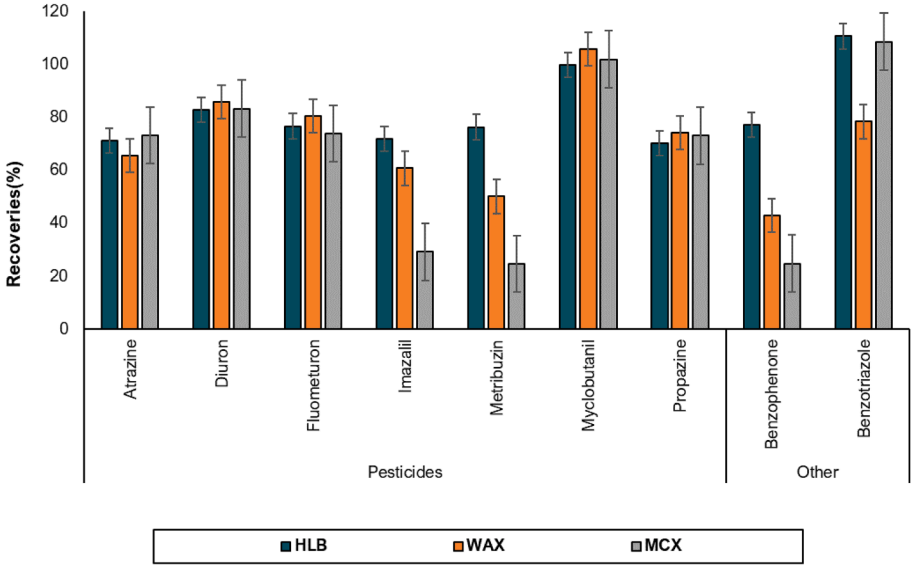


Fig. 1. Sample extraction workflow for (a) HLB, (b) WAX, (C) MCX cartridges.



(a)



(b)

Fig. 2. Recoveries of studied PMOCs(a) PPCPs and (b) pesticides after the application of different SPE protocols.

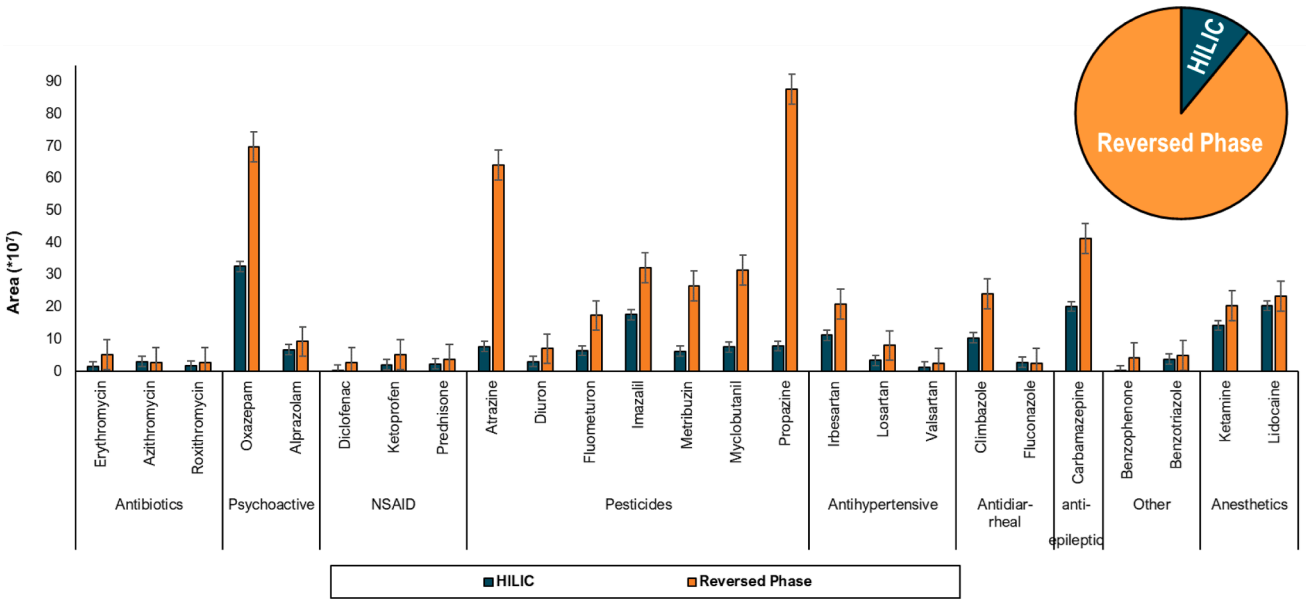


Fig. 3. Sensitivity enhancements (in terms of intensity) by changing the column.

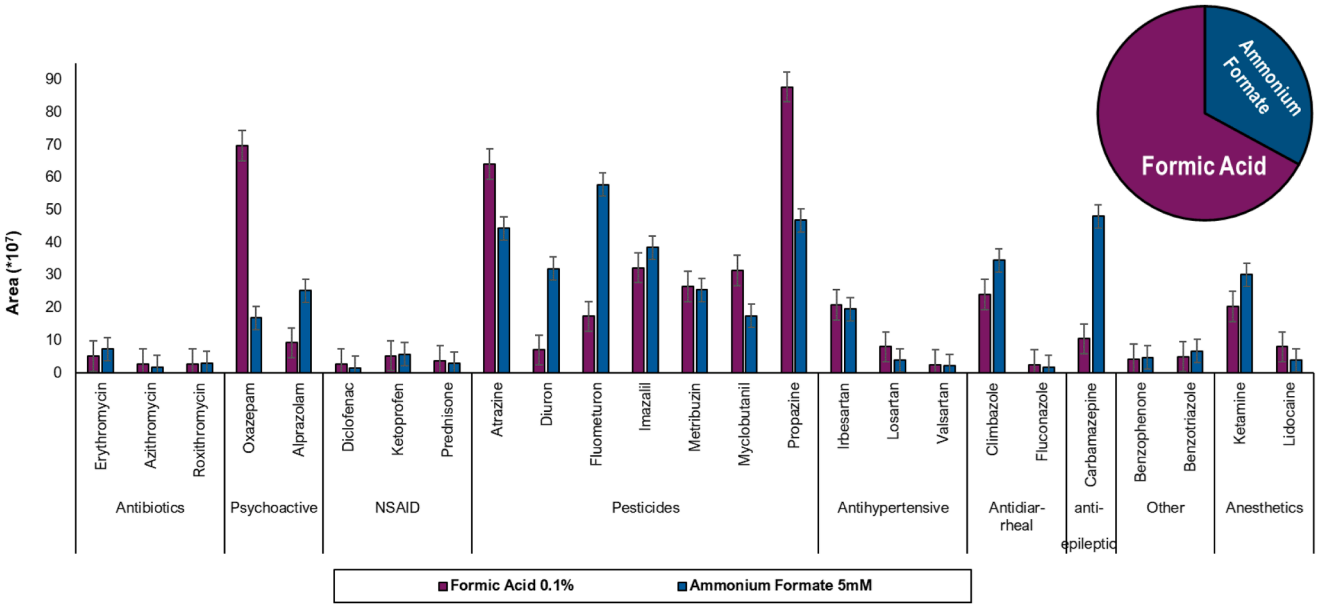


Fig. 4. Sensitivity enhancements (in terms of intensity) by changing the column.

3.2. Evaluation of critical chromatographic conditions

3.2.1. Column type

To examine the crucial factors influencing the chromatography, the procedure was refined with regard to the working solutions' capacity for separation, peak shape, and lack of carryover effects. This was accomplished by using a working solution containing every examined compound at a 50 ng/L concentration. An acidified water–methanol mobile phase was used for the RP column and 10 mM ammonium acetate in water/acetonitrile (90/10, v/v) and 10 mM ammonium acetate in acetonitrile/water (10/90, v/v) for HILIC column. The two evaluated columns were: (i) Thermo Hypersil GOLD aQ column (50 × 2.1 mm, 1.9 μm) and (ii) HILIC (150 mm × 2.1 mm, 2.7 μm).

The polarity and chemical characteristics of each molecule determine whether a hydrophilic interaction liquid chromatography (HILIC) column or a reversed-phase (RP) column will be the best choice for retention and resolution. RP columns are ideal for the analysis of non-polar to moderately polar compounds due to their retention mechanism which is based on hydrophobic interactions between the non-polar parts of the analyte and the non-polar stationary phase. On the other hand, HILIC columns are suitable for the analysis of polar and hydrophilic compounds owing to their retention mechanism, which is based on hydrophilic interactions between the polar parts of the analyte and the hydrophilic stationary phase. As illustrated in Fig. 3, the results from both columns were satisfactory, while the Hypersil GOLD aQ column performed the best in terms of signal intensity and providing better peak characteristics concerning the separation, analysis rate, and sharp peaks. The high efficiency of this column can be attributed to its ability to enhance interactions such as π-π, dipole–dipole, and ion-exchange interactions between the analytes and stationary phase, aided by the presence of water molecules in the mobile phase. However, for some very polar compounds (see Table S1 with physicochemical properties)

like cetirizine, fluconazole, oxolinic acid and sulfadiazine, the HILIC column seems to be a better option. Due to its compatibility with the authors' previously published multi-residue approach, the Hypersil GOLD aQ column was chosen for PMOC analysis. This eliminates the need to switch columns and run samples twice.

3.2.2. Mobile phase composition

The following step involved optimizing the mobile phase after establishing the suitable column. In particular, the following mixtures were investigated: (i) (A) water + 0.1 % FA and (B) methanol + 0.1 % FA; (ii) (A) water + 0.1 % FA and (B) acetonitrile + 0.1 % FA (iii) (A) water + 0.1 % AA and (B) methanol + 0.1 % AA and (iv) (A) water (5 mM NH_4F) and (B) methanol (5 mM NH_4F). The use of acetonitrile marginally improved the peak shape of the compounds eluting at the end of the chromatogram when compared to other solvents. Since methanol provided more than satisfactory chromatographic performance, acetonitrile was rejected to prevent signal suppression and lower the cost of the analysis. Regarding the addition of acid to the mobile phase, it is commonly known that this alteration enhances most compounds' responses by producing H^+ ions. Furthermore, because $H+$ is more abundant, the synthesis of the undesired $[M + Na]^+$ ions is inhibited or limited . Since the pH of the solution was higher, there was no discernible change in the separation and peak shape when acetic acid was used instead of formic acid. Finally, the comparison between formic acid and ammonium formate showed minimal differences in signal intensity for most compounds (Fig. 4). According to the literature, the addition of formic acid exhibits less signal suppression [17]. What is more, the use of organic salts can compromise the long-term stability of the Orbitrap system, necessitating a thorough flush that makes the entire procedure time-consuming and complex. As a result, the inclusion of formic acid was found to be the optimal compromise for PMOC chromatography and system integrity based on the aforementioned information.

3.3. Central composite design – MS optimization

Finally, in the present study, a central composite design in mass spectrometry was performed. As mentioned above, the studied parameters were the probe position, the capillary and auxiliary temperature. To begin with, the auxiliary temperature is essential for the effective dissolution of analyte droplets. Proper desolation of the ions produces a higher-quality and more intense signal, ensuring their efficient transmission to the mass spectrometer. Thus, auxiliary temperature is a key parameter for method optimization because it can be adjusted to balance the thermal stability of analytes and method sensitivity. Achieving high sensitivity requires sufficient evaporation of solvent molecules, which can be accomplished with the correct auxiliary temperature, thereby considerably improving ionization efficiency. The aim is to determine the optimal auxiliary temperature for ionization efficiency while avoiding undue thermal breakdown or analyte instability. Lower auxiliary temperatures might be required for thermally sensitive chemicals to prevent deterioration, while higher auxiliary temperatures might be helpful for effectively dissolving solvents with high boiling points. The studied values of auxiliary temperature that were chosen for optimization were 200, 300 and 400 °C based on the literature [18].

Moving on to the second parameter, in an LC-Orbitrap HRMS system, the capillary temperature is a crucial variable that can have significant impact on the ionization process and, in turn, the accuracy of the mass spectrometry data. In the case of ESI, a capillary at atmospheric pressure with an applied electric field of a few kV introduces the chemicals diluted in the mobile phase. There are competing theories about the ionization process, including the residual charge theory proposed by Dole et al. in 1968, and the ion evaporation model [19]. In detail, the analyte's desolation as it enters the mass spectrometer is influenced by the capillary temperature. An enhanced solvent evaporation process can result in more effective analyte ionization at higher temperatures. Higher temperatures can enhance solvent evaporation and improve analyte ionization; however, excessive temperatures can lead to the thermal decomposition of sensitive analytes. Having the right capillary temperature can reduce the effects of impurities or leftover solvents on ion suppression. It also contributes to clearer spectra by lowering background noise. Based on the literature, the investigated capillary temperature values of 280, 300, and 320 °C were selected for optimization [20].

Finally, the last studied parameter was the probe position, which significantly influences the efficiency of analyte ionization. Optimal probe positioning ensures that the spray is directed correctly into the ionization region, maximizing ion production. Optimal probe positioning ensures that the spray is directed appropriately into the ionization region, maximizing ion production and signal strength. When the probe is positioned correctly, more ions are produced and transmitted into the mass analyzer, which can result in a stronger signal. A closer probe position may lead to more efficient ionization as analytes are ionized closer to the mass spectrometer's entry. A shorter distance can create a more concentrated ion beam, reducing ion loss from dispersion or scattering. However, being nearer to the source of ionization can raise the risk of contamination from matrix effects or solvent droplets and might also expose the probe to higher temperatures, which could affect thermally labile analytes. On the other hand, a greater distance can aid in lowering contamination from matrix elements and solvent droplets, producing cleaner spectra. Still, longer-distance ions might be more prone to dispersal and scattering, which could result in a weaker signal.

According to Reymond et al., 2019 compounds' signal regions for ESI may even be five times lower depending on the location of the ESI sprayer [20]. In an ESI source, ionization occurs due to the high voltage between the ESI needle and the ion sweep cone, which creates a spray in the liquid phase. Hence, the electric field (E) is directly influenced by the location of the ESI sprayer and increases in intensity with sprayer proximity Eq. (1).

$$\mathrm{E} = \frac{\boldsymbol{V}}{\boldsymbol{A}\cdot \boldsymbol{r}\cdot \ln\left(\frac{4d}{r}\right)} \tag{1}$$

where V is the applied voltage, r is the capillary emitter radius (in centimeters), d is the distance (in centimeters) between the ESI sprayer and the ground plane, and A is an empirical constant that Smith determined to be equal to 0.667 [21].

Consequently, an overall desirability function was generated, ranging from 0 (least probability) to 1 (highest probability) of goal achievement (Design-Expert®, 2009). As can be seen from the above equation, the model with the best fit to the experimental data was the quadratic model Eq. (2).

$$\begin{aligned}\mathbf{Y} = {} & +\mathbf{1.986}*\mathbf{10^8}-\mathbf{5.59}*\mathbf{10^6}*\mathbf{A}-\mathbf{1.60610^6}*\mathbf{B}+\mathbf{5.378}*\mathbf{10^5}*\mathbf{C[1]}-\mathbf{1.149}*\mathbf{10^7}*\mathbf{C[2]}-\mathbf{1.522}*\mathbf{10^7}*\mathbf{A}*\mathbf{B}+\\ & \mathbf{2.209}*\mathbf{10^7}*\mathbf{A}*\mathbf{C[1]}-\mathbf{1.218}*\mathbf{10^7}*\mathbf{A}*\mathbf{C[2]}--\mathbf{1.514}*\mathbf{10^7}*\mathbf{B}*\mathbf{C[1]}+\mathbf{1.363}*\mathbf{10^7}*\mathbf{B}*\mathbf{C[2]}--\mathbf{2.786}*\mathbf{10^7}*\mathbf{A^2}+\\ & \mathbf{8.715}*\mathbf{10^6}*\mathbf{B^2}--\mathbf{1.379}*\mathbf{10^6}*\mathbf{A}*\mathbf{B}*\mathbf{C[1]}+\mathbf{5.658}*\mathbf{10^6}*\mathbf{A}*\mathbf{B}*\mathbf{C[2]}--\mathbf{1.379}*\mathbf{10^7}*\mathbf{A^2C[1]}+\mathbf{2.369}*\mathbf{10^6}*\mathbf{A^2C[2]}+\\ & \mathbf{5.590}*\mathbf{10^7}*\mathbf{B^2C[1]}-\mathbf{2.930}*\mathbf{10^5}*\mathbf{B^2C[2]}\end{aligned} \tag{2}$$

Table 1
Estimation of significance of parameters by ANOVA analysis.

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	4,12E+19	17	2,42E+18	15.48	< 0.0001	significant
A-Auxillary temperature	5,63E+17	1	5,63E+17	3.60	0.0772	
B-Capillary temperature	4,64E+16	1	4,64E+16	0.2965	0.5941	
C-probe	8,98E+18	2	4,49E+18	28.68	< 0.0001	
AB	2,78E+18	1	2,78E+18	17.76	0.0008	
AC	4,41E+18	2	2,20E+18	14.08	0.0004	
BC	2,50E+18	2	1,25E+18	8.00	0.0043	
A^2	5,90E+18	1	5,90E+18	37.69	< 0.0001	
B^2	5,77E+17	1	5,77E+17	3.69	0.0740	
ABC	2,09E+17	2	1,05E+17	0.6676	0.5276	
A^2C	8,81E+17	2	4,41E+17	2.82	0.0916	
B^2C	1,58E+19	2	7,87E+18	50.32	< 0.0001	
Residual	2,35E+18	15	1,57E+17			
Lack of Fit	1,06E+18	9	1,17E+17	0.5441	0.8023	not significant
Pure Error	1,29E+18	6	2,15E+17			
Cor Total	4,35E+19	32				

In order to confirm the significance of the model, both the fit to experimental data and prediction of the response were assessed. The significance of the model is confirmed, since the p-value is 0.001< 0.05, resulting in an acceptable correlation between the experimental and predicted values. The above significance is further confirmed by the F value calculated equal to 15.48. Furthermore. the probability of the F value being so small due to noise is 0.01 %. The lack of fit was found to be non-significant, with a p value equal to 0.5441 and an F value equal to 0.8023 (Table 1), indicating that the probability of model failure is small to negligible.

In addition, the quality of the fitted quadratic polynomial model and its overall predictive ability was further evaluated by the coefficient of determination (R^2), adjusted R^2 (Adj. R^2) and CV. The R^2 was 0.9461, indicating that 94.61 % of the variance is associated with the independent variables and can be specified by the model with a CV of 6.65 %. The statistical precision measure Adj. $R^2 = 0.8850$, which is also high to support the significance of the model, suggests that 88.50 % of the data points in the estimated regression line are explained only by the independent variables actually affecting the dependent variable, but adjusted for the number of model terms. The predicted R^2 of 0.8850 is in reasonable agreement with the adjusted R^2 of 0.7008 (the difference is less than 0.2).

A total of 33 experiments were conducted, each of which included three focal points, with three replicates per experiment and three replicates for each focal point. The experiments were conducted in random order to account for possible uncontrolled variables, with the aim of reducing their effect on the effectiveness of the method. The variables and levels tested were determined based on previous research [19,20,22,23]. Multiple linear regressions were used to derive the corresponding polynomial models. Table 2 presents the main factors, symbols, levels and responses of each run.

To confirm the suitability of the model, further diagnostic tests were performed, and several charts were drawn. In Fig. 5(i), the Box-Cox diagnostic test for power transformations showed that the current value, equal to 1, is within the acceptable range (−2 to 3), indicating an acceptable result. Furthermore, Fig. 5(ii), showing the Normal Plot of Residuals, shows a good linear correlation of the responses. Finally, Fig. 5 (iii), showing the correlation of the predicted responses with the actual responses, shows that most points are close to the line, indicating that the systematic error does not exceed the threshold value of ±3.

Different combinations of two factors interact each time with the average area to visualize the relationship between the response and the experimental levels of the factors. This interaction is illustrated through contour plots. In particular, Figure 5(iv), (v), (vi) shows that the average peak area increases with decreasing distance from the source (positions B, C, D). Position B seems to be the optimal position for the source.

The desirability function (D) was also used as a multiple response criterion to achieve a better analytical response. A function for each analytical response is generated through a process that results in a global function D. In the end, this function must be maximized by selecting the optimal conditions of the variables for the optimal analytical response of the analytes under study. The numerical optimization of the software was defined by assigning predicted values on a scale from 0.0 (undesirable) to 1.0 (highly desirable). The optimal conditions thus obtained were as follows: A: auxillary temperature = 318.515 °C, B: capillary temperature = 280.803 °C and C: probe position = *B*.

According to manufacturer recommendations [24], several additional parameters, including sheath gas flow, sweep gas flow, and auxiliary gas flow rates, are frequently well-optimized in conventional instrument designs. Changing these parameters might not result in appreciable gains without thorough re-optimization research. Concentrating on too many parameters at once can make optimization more difficult. A more controlled and methodical study with fewer variables will guarantee more reliable results. Time, sample availability, and analytical resources may limit the scope of the optimization study. Prioritizing the most impactful factors enables effective use of available resources.

Factor	Name	Type	SubType	Minimum	Maximum	Coded Low	Codes High	Mean	Std. Dev.
A	Auxillary temperature	Numeric	Continuous	200	400	−1 ↔ 200.00	+1 ↔ 400.00	300	75
B	Capillary temperature	Numeric	Continuous	280	320	−1 ↔ 280.00	+1 ↔ 320.00	300	15
C	probe	Categoric	Nominal	B	D			Levels:	3

Based on the results of the ANOVA analysis for the central composite design, the combination of parameters A: auxillary temperature and B: capillary temperature was significant, with a p-value of 0.0008 (p-value <0.05). In contrast, the combination of all three parameters (A, B and C) was not considered significant, since the p-value was 0.6676, greater than 0.05.

Table 2
The 33 CCD experiments with factors and study levels.

Std	Run	Factor 1: A: Aux. Heater Temp.	Factor 2: B: Capillary Temp.	Factor 3: C: Probe Position	Response 1 Area Average
3	1	200	320	B	201,107,000
11	2	300	300	B	215,224,818
27	3	200	300	D	216,973,713
8	4	300	320	B	245,488,000
13	5	400	280	C	145,714,942
4	6	400	320	B	210,296,000
10	7	300	300	B	210,783,000
14	8	200	320	C	216,445,000
28	9	400	300	D	165,391,014
16	10	200	300	C	175,630,803
5	11	200	300	B	147,169,000
26	12	400	320	D	117,119,000
25	13	200	320	D	176,828,322
17	14	400	300	C	141,746,020
33	15	300	300	D	219,226,378
6	16	400	300	B	161,347,615
30	17	300	320	D	161,711,755
32	18	300	300	D	196,132,400
2	19	400	280	B	278,278,147
7	20	300	280	B	276,387,514
31	21	300	300	D	218,015,969
23	22	200	280	D	139,539,392
19	23	300	320	C	194,331,035
15	24	400	320	C	160,948,123
24	25	400	280	D	157,822,000
21	26	300	300	C	181,696,045
9	27	300	300	B	177,047,054
20	28	300	300	C	193,483,624
1	29	200	280	B	202,696,000
22	30	300	300	C	192,044,743
18	31	300	280	C	190,863,530
29	32	300	280	D	158,903,041
12	33	200	280	C	162,969,875

3.4. Method performance

The developed analytical method was validated using tap water spiked at a specific "level of interest" concentration of 50 ng/L. The validation tests were conducted independently on various days. The precision and accuracy of the technique, linearity (coefficients of determination), ME, and method detection and quantification limits (MDLs, MQLs) were among its performance attributes. Table S1 of the Supplementary Material lists all the analytical attributes.

Compounds that were not strongly retained by the HLB adsorbents, such as isoniazid, tamoxifen, metformin, sertraline, and cimetidine, exhibited the lowest recoveries (< 30 %). Of the chemicals under study, 20 % were recovered within the still-acceptable range of 30–70 %, while 60 % were recovered within the range of 70–120 %. Merely 20 % of the target analytes exhibited low recovery (<30 %).

The lowest concentration of the analyte for which the S/N ratio (determined with the help of the RMS algorithm) surpassed 10 following injections into the column was used to measure the instrumental sensitivity. Analyte concentrations corresponding to S/N ratios of 3.3 and 10, respectively, were calculated after the peak met all identification criteria. Similarly, water samples were spiked at low levels and subjected to SPE under optimal conditions to determine the method's detection and quantification limits. Most of the compounds had MQLs < 1 ng/L or even < 0.5 ng/L. However, several compounds had higher MQLs, namely antibiotics, which have previously been documented elsewhere [25–27]. These higher limits are primarily due to poor retention of these compounds on the sorbent. Given the careful optimization during sample preparation, the computed limits are generally in good agreement with those stated in pertinent literature, if not lower.

4. Conclusions and perspectives

In the present study, a comprehensive approach was employed to develop, optimize, and validate a new analytical method for the detection of PMOCs in aqueous samples using Liquid Chromatography-High Resolution Mass Spectrometry (LC-HRMS). The challenges associated with PMOC analysis, particularly their diverse physicochemical properties, were addressed through the systematic optimization of both sample preparation and instrumental analysis.

The first step focused on optimizing solid-phase extraction. The study compared different SPE cartridges—Hydrophilic-Lipophilic Balance (HLB), Weak Anion Exchange (WAX), and Mixed-Mode Cation Exchange (MCX)—to enhance the recovery of PMOCs. Each cartridge displayed varying extraction efficiencies depending on the chemical nature of the analytes. HLB cartridges provided the most consistent recoveries (70–120 %) for a broad range of compounds, whereas MCX cartridges were particularly effective for basic compounds, and WAX cartridges excelled in the extraction of acidic substances. These results underscore the importance of selecting the appropriate SPE material based on the specific properties of target compounds. The second phase of the study focused on method sensitivity and reproducibility. The method demonstrated excellent linearity ($r^2 > 0.997$) for all compounds, indicating its capacity to accurately quantify low concentrations of PMOCs in water. The recoveries of most analytes fell within the acceptable range according to ISO/IEC 17,025:2017 standards, ensuring that the method is reliable for environmental analysis. Precision was also strong, with relative standard deviations (RSD) within the accepted limits, affirming the reproducibility of the method across multiple days and sample replicates.

What is more, the study explored two chromatographic methods using both HILIC and RPLC columns. While both columns performed satisfactorily, the Hypersil GOLD aQ column was selected due to its superior separation, peak shape, and compatibility with the broad range of PMOCs. In mobile phase selection and optimization step, methanol was chosen over acetonitrile to avoid signal suppression, and formic acid was found to enhance analyte ionization without causing significant signal suppression or instrument contamination. Finally, a central composite design (CCD) approach was employed to optimize mass spectrometric parameters, such as capillary temperature, auxiliary temperature, and probe position. This statistical approach allowed for a fine-tuning of the HRMS conditions, significantly improving signal intensity and overall method sensitivity. The optimization process revealed that an auxiliary temperature of approximately 318 °C, capillary temperature of around 280 °C, and probe position "B" provided the best ionization efficiency and analyte response. In conclusion, this optimized SPE-LC-HRMS method offers a sensitive, robust, and cost-effective solution for the comprehensive analysis of PMOCs in water. It addresses the limitations of traditional analytical methods by providing high recoveries for a wide range of analytes, reducing sample preparation time, and improving method efficiency. This approach enhances the ability to monitor and assess the presence of persistent pollutants in water bodies, contributing to the protection of public health and environmental safety.

Funding

This research was funded by European Commission under the HORIZON-WIDERA- TWINNING BOTTOM-UP within the framework of the SPECTRA project (Grant No 101,158,453).

CRediT authorship contribution statement

Kyriaki Anagnostopoulou: Writing – original draft, Investigation, Formal analysis, Data curation. **Vasileios Alampanos:** Writing – original draft, Visualization, Methodology, Data curation. **Eleni Evgenidou:** Writing – review & editing, Visualization, Methodology, Formal

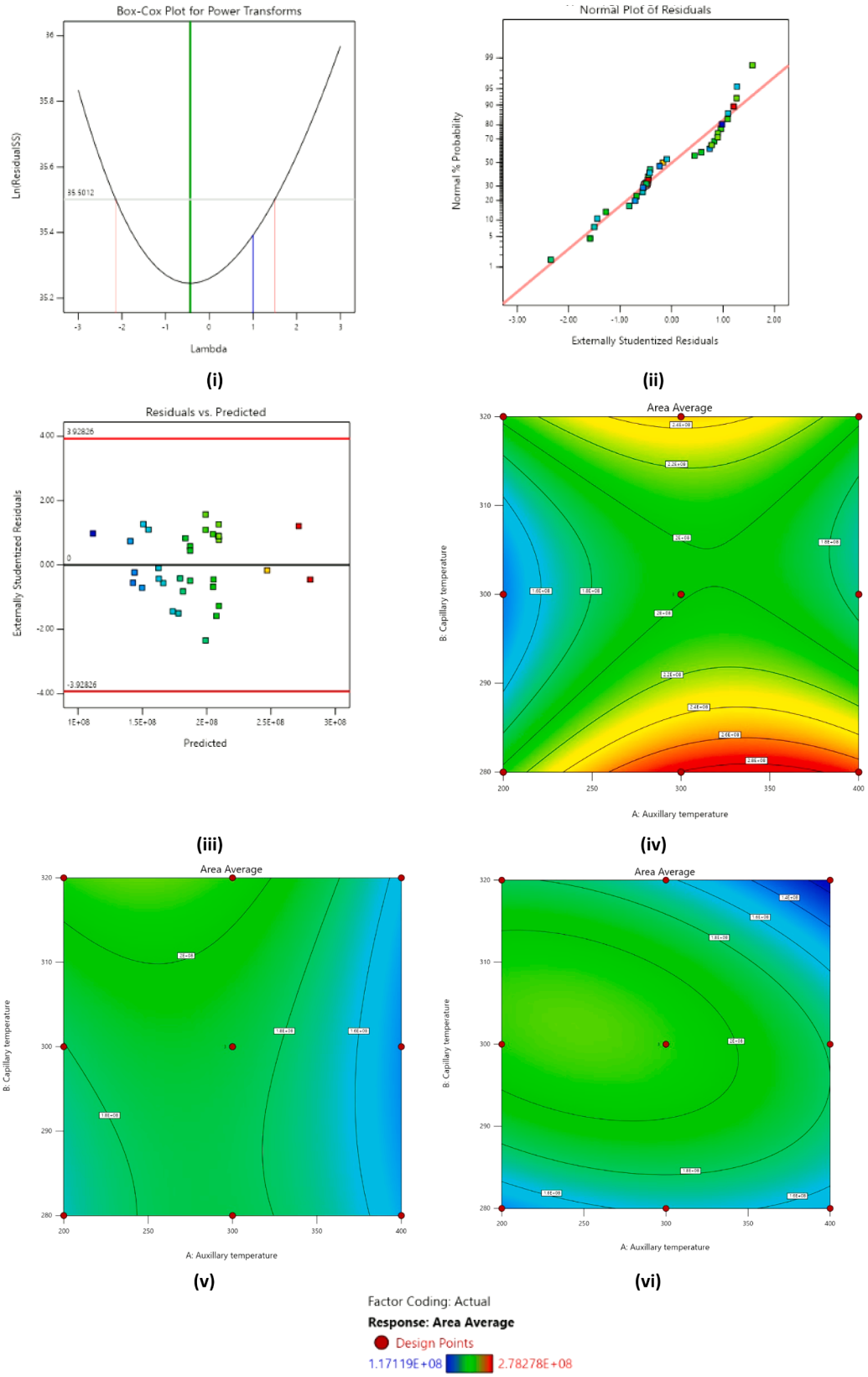


Fig. 5. Graph derived from the central composite design (i) Normal Plot of Residuals (ii) Box-Cox diagnostic test (iii), correlation of predicted responses with actual responses, (iv) contour plot, correlation of A and B at position B, (v) contour plot, correlation of A and B at position C, (vi) contour plot, correlation of A and B at position D.

analysis. **Dimitra A. Lambropoulou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research work was supported by the Hellenic Foundation for Research and Innovation (HFRI) under the 4th Call for HFRI PhD Fellowships (Fellowship Number: 11182).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.greeac.2025.100214.

Data availability

The authors are unable or have chosen not to specify which data has been used.

References

[1] H.P.H. Arp, T.N. Brown, U. Berger, S.E. Hale, Ranking REACH registered neutral, ionizable and ionic organic chemicals based on their aquatic persistency and mobility, Environ. Sci. Process. Impacts 19 (2017) 939–955, https://doi.org/10.1039/c7em00158d.
[2] P. Chirsir, E.H. Palm, S. Baskaran, E.L. Schymanski, Z. Wang, R. Wolf, S.E. Hale, H. P.H. Arp, Grouping strategies for assessing and managing persistent and mobile substances, Technology (2024), https://doi.org/10.1186/s12302-024-00919-4.
[3] H.P.H. Arp, S.E. Hale, REACH: improvement of guidance and methods for the identification and assessment of PMT/vPvM substances, Natl. Audit Violence (2003 - 2005) (2019) 131.
[4] T. Mohr, I. Schliebner, M. Neumann, L. Oules, H.P.H. Arp, S.E. Hale, Progress in European chemicals policy to support the protection of the environment and human health from persistent, mobile and toxic and very persistent and very mobile substances, Environ. Sci. Eur. 36 (2024), https://doi.org/10.1186/s12302-024-00932-7.
[5] European Commission, Commission delegated regulation (EU) 2023/707, 2016 (2022) 48–119.
[6] R. Montes, R. Rodil, R. Cela, J.B. Quintana, Determination of persistent and mobile organic contaminants (PMOCs) in water by mixed-mode liquid chromatography-tandem mass spectrometry, Anal. Chem. 91 (2019) 5176–5183, https://doi.org/10.1021/acs.analchem.8b05792.
[7] I. Neuwald, M. Muschket, D. Zahn, U. Berger, B. Seiwert, T. Meier, J. Kuckelkorn, C. Strobel, T.P. Knepper, T. Reemtsma, Filling the knowledge gap: a suspect screening study for 1310 potentially persistent and mobile chemicals with SFC- and HILIC-HRMS in two German river systems, Water Res 204 (2021) 117645, https://doi.org/10.1016/j.watres.2021.117645.
[8] S. Minkus, S. Grosse, S. Bieber, S. Veloutsou, T. Letzel, Optimized hidden target screening for very polar molecules in surface waters including a compound database inquiry, Anal. Bioanal. Chem. 412 (2020) 4953–4966, https://doi.org/10.1007/s00216-020-02743-0.
[9] M.C. Hennion, Solid-phase extraction: method development, sorbents, and coupling with liquid chromatography, J. Chromatogr. A 856 (1999) 3–54, https://doi.org/10.1016/S0021-9673(99)00832-8.
[10] N. Gilart, F. Borrull, N. Fontanals, R.M. Marcé, Selective materials for solid-phase extraction in environmental analysis, Trends Environ. Anal. Chem. 1 (2014), https://doi.org/10.1016/j.teac.2013.11.002.
[11] S. Lacorte, C. Molina, D. Barceló, Screening of organophosphorus pesticides in environmental matrices by various gas chromatographic techniques, Anal. Chim. Acta 281 (1993) 71–84, https://doi.org/10.1016/0003-2670(93)85341-G.
[12] Waters, OASIS ® MCX product and generic method information, (2020) 12.
[13] A. Chabilan, N. Landwehr, H. Horn, E. Borowska, Impact of log(Kow) value on the extraction of antibiotics from river sediments with pressurized liquid extraction, Water (Switzerland) 14 (2022) 1–17, https://doi.org/10.3390/w14162534.
[14] G. Hodges, C. Eadsforth, B. Bossuyt, A. Bouvy, M.H. Enrici, M. Geurts, M. Kotthoff, E. Michie, D. Miller, J. Müller, G. Oetter, J. Roberts, D. Schowanek, P. Sun, J. Venzmer, A comparison of log K ow (n-octanol–water partition coefficient) values for non-ionic, anionic, cationic and amphoteric surfactants determined using predictions and experimental methods, Environ. Sci. Eur. 31 (2019) 1–18, https://doi.org/10.1186/s12302-018-0176-7.
[15] European Comission, Document n0 SANTE 11312/2021. Analytical quality control and method validation procedures for pesticide residues analysis in food and feed., (2021) 1–57.
[16] S. Yang, R.E. Synovec, M.G. Kalyuzhnaya, M.E. Lidstrom, Development of a solid phase extraction protocol coupled with liquid chromatography mass spectrometry to analyze central carbon metabolites in lake sediment microcosms, J. Sep. Sci. 34 (2011) 3597–3605, https://doi.org/10.1002/jssc.201100533.
[17] D. Johnson, B. Boyes, R. Orlando, The use of ammonium formate as a mobile-phase modifier for LC-MS/MS analysis of tryptic digests, J. Biomol. Tech. 24 (2013) 187–197, https://doi.org/10.7171/jbt.13-2404-005.
[18] I.M. Huérfano Barco, J.C. España Amórtegui, J.A. Guerrero Dallos, Development and validation of qualitative screening, quantitative determination and post-targeted pesticide analysis in tropical fruits and vegetables by LC-HRMS, Food Chem 367 (2022), https://doi.org/10.1016/j.foodchem.2021.130714.
[19] N. Lemonakis, A.L. Skaltsounis, A. Tsarbopoulos, E. Gikas, Optimization of parameters affecting signal intensity in an LTQ-orbitrap in negative ion mode: a design of experiments approach, Talanta 147 (2016) 402–409, https://doi.org/10.1016/j.talanta.2015.10.009.
[20] C. Reymond, A. Le Masle, C. Colas, N. Charon, A rational strategy based on experimental designs to optimize parameters of a liquid chromatography-mass spectrometry analysis of complex matrices, Talanta 205 (2019) 120063, https://doi.org/10.1016/j.talanta.2019.06.063.
[21] D.P.H. Smith, The electrohydrodynamic atomization of liquids, IEEE IA-22 (1986) 527–535, https://doi.org/10.1109/TIA.1986.4504754.
[22] E.S. Hecht, A.L. Oberg, D.C. Muddiman, Optimizing mass spectrometry analyses: a tailored review on the utility of design of experiments, J. Am. Soc. Mass Spectrom. 27 (2016) 767–785, https://doi.org/10.1007/s13361-016-1344-x.
[23] L. Hu, Y. Li, W. Zhang, Characterization and application of surface-molecularimprinted-polymer modified TiO2 nanotubes for removal of perfluorinated chemicals, Water Sci. Technol. 74 (2016) 1417–1425, https://doi.org/10.2166/wst.2016.321.
[24] Thermo Scientific, TSQ series manual (2010).
[25] A. Ofrydopoulou, C. Nannou, E. Evgenidou, A. Christodoulou, D. Lambropoulou, Assessment of a wide array of organic micropollutants of emerging concern in wastewater treatment plants in Greece: occurrence, removals, mass loading and potential risks, Sci. Total Environ. 802 (2022) 149860, https://doi.org/10.1016/j.scitotenv.2021.149860.
[26] J. Robles-Molina, F.J. Lara-Ortega, B. Gilbert-López, J.F. García-Reyes, A. Molina-Díaz, Multi-residue method for the determination of over 400 priority and emerging pollutants in water and wastewater by solid-phase extraction and liquid chromatography-time-of-flight mass spectrometry, J. Chromatogr. A 1350 (2014) 30–43, https://doi.org/10.1016/j.chroma.2014.05.003.
[27] K. Anagnostopoulou, E. Evgenidou, D.A. Lambropoulou, Advanced suspect screening of tiamulin and its transformation products in waters : assessing their persistence, mobility and toxicity, J. Environ. Chem. Eng. 12 (2024) 112392, https://doi.org/10.1016/j.jece.2024.112392.